

The Borodin–Birkhoff intertwiner

Graph resolvent of the source-pure min-norm extension, and why Φ_N is a Woodbury template, not the terminal innovation

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Abstract

Rounds 404 and 405 compared Euler/Digamma Grams to $A_0 = R_{N-3} - \frac{1}{2}I$ without a dual quotient (residual of order 209). Round 407 wrote the dictionary $R = C(I + C)^{-1}$ with $C = \mathfrak{T}^*\mathfrak{T}$ in the dual orthogonal-polynomial basis (FRAME-A residual $1.8 \cdot 10^{-14}$). This note lifts that dictionary to a Jacobi-free source formula for the operator $\mathfrak{T}_0$ itself: the min-norm interpolant $Y \rightarrow X$ in $\mathcal{P}_{<d_0}$, $d_0 = N_w - 3$. Then $R_{\text{graph}} = (I + \mathfrak{T}_0^*\mathfrak{T}_0)^{-1}$ equals the r356 dual resolvent, and $\text{ind}_-(I - \mathfrak{T}_0^*\mathfrak{T}_0) = \text{ind}_- A_0 = \text{ind}_- \mathcal{D}_0$. The r369 Haynsworth matrix $\Phi_N = J^{-1} + U^T A_0^{-1} U$ is the Woodbury precision template (r406 Lean `woodbury_inv`); it is *not* the pair-plus-border Schur of the augmented Gram, so the candidate $\Theta_{\text{terminal}} = C^T \Phi_N C$ is **refuted** as a literal matrix identity. Sequential Uvarov innovations satisfy $n_{\text{neg}}^{\text{inn}} \leq 1$ on core-42 (census, 42/42), not a theorem, and do not identically equal $\text{ind}_- A_0$ (32/42). The unaugmented operator on six terminal-dead χ windows stays near contractive; the r362 graph factor $\mathfrak{T}^\dagger$ of the genuine border functional flips all six ($\|\mathfrak{T}^\dagger\| > 1$). That last identity uses the r362 Gram: it is a dictionary, not a new theorem of $R^\dagger$. No RH claim.

Status. Lemmas 1–2 are proved (finite identities over $\mathbb{Q}$). Lemmas 3–4 and Propositions 1–3 are a finite machine census on FRAME-A, core-42, named MAIN windows, and the χ twins. The literal identification $\Theta = C^T \Phi_N C$ is **refuted**. Unfitted Krein transport of Euler/Digamma is **refuted**. Ausgang ONE_DEFECT_TRANSPORT_CENSUS. No RH claim. No L^* claim. No $R^\dagger \succ \frac{1}{2}I$ claim.

1 The source formula

Write u for the folded signed union weight on the cosine grid, $Y = \text{supp}(u_-)$ the hole nodes, X the μ -nodes, and $u_j^\vee = 1/(u_j P_{\Xi'}^2(x_j))$ the Borodin dual (r356; $P_{\Xi'}$ the node polynomial of the grid). Let $d_0 = N_w - 3$, $W_X = \text{diag}(u^\vee|_X)$, $W_Y = \text{diag}(u^\vee|_Y)$. Let $L_{ij} = \ell_j(x_i)$ be Lagrange on Y evaluated on X , and let $K_{ia} = (P_Y \cdot \varphi_a)(x_i)$ for a basis $(\varphi_a)_{a < d_0 - |Y|}$ of $\mathcal{P}_{<d_0 - |Y|}$. The weighted kernel projector is

$$\Pi_K = W_X^{1/2} K (K^T W_X K)^{-1} K^T W_X^{1/2}. \quad (1)$$

The min-norm interpolant in $\mathcal{P}_{<d_0}$ is

$$\mathfrak{T}_0 = (I - \Pi_K) W_X^{1/2} L W_Y^{-1/2}. \quad (2)$$

Equivalently, the KKT of constrained least squares $\min \|W_X^{1/2} \Phi_X c\|$ subject to $\Phi_Y c = v$. Lagrange-plus- Π_K is $\mathbb{Q}$ -exact and catastrophically ill-conditioned at working depth; the stable evaluation is Chebyshev CLS. No R , A_0 , eigenvector, or target Cholesky enters the constructor.

The Schur short-circuit Gram on Y is $S_0 = L^T W_X L - L^T W_X K (K^T W_X K)^{-1} K^T W_X L$, and $\mathcal{D}_0 = W_Y - S_0$ is congruent to $I - \mathfrak{T}_0^* \mathfrak{T}_0$ (Haynsworth I7).

Lemma 1 (Three bases, kernel, graph over $\mathbb{Q}$). *On the five-node toy $\{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$ with positive square masses $\{4, 9, 1, 16, 25\}$, holes $Y = \{0, \frac{1}{2}\}$, and $d_0 = 3$, the dual masses $u^\vee = 1/(uP'^2)$ are rational squares. Monomial, Newton, and Chebyshev bases produce identical S_0 ; $\ker(C_Y)$ has dimension $d_0 - |Y| = 1$; and the dressed resolvent $W_Y^{1/2}(W_Y + S_0)^{-1}W_Y^{1/2}$ equals both the dual Christoffel–Darboux Gram of depth d_0 and $C(I + C)^{-1}$ with C the μ -only dual Gram on Y . The mutant $u^\vee = 1/u$ (no P'^2) and the shifts $d_0 \pm 1$ break the identity.*

Lemma 2 (Woodbury rank-one precision). *Over $\mathbb{Q}$, for $H = \begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}$, $e = (1, -1)$, $\omega = 2$,*

$$\eta = e^T H^{-1} e = \frac{9}{11}, \quad \Theta = (\omega^{-1} + \eta)^{-1} = \frac{22}{29},$$

and $S^+ - S = \Theta e e^T$. This is the scalar case of $S^+ - S = E^T(\Omega^{-1} + K_0 A^{-1} K_0^T)^{-1} E$ with $E = L_0 - K_0 A^{-1} B$ (r406 woodbury_inv / oneDefectDelta_eq_woodbury, sorry-free).

Lemma 3 (Machine graph identity). *On FRAME-A, Chebyshev CLS at $d_0 = N_w - 3$ yields $\|R_{\text{graph}} - R_{\text{chain}}\|_F / \|R\|_F = 2.16 \cdot 10^{-14}$ against the independent r407 chain builder, and $\mathfrak{T}_0^* \mathfrak{T}_0 = R^{-1} - I$ to $7.47 \cdot 10^{-14}$. $\|\mathfrak{T}_0\|_2 = 1.08014$ (exactly one singular value > 1). $\text{ind}_-(I - \mathfrak{T}_0^* \mathfrak{T}_0) = \text{ind}_- A_0 = \text{ind}_- \mathcal{D}_0 = 1$. Named MAIN windows: $k_z = 15$ residual $3.18 \cdot 10^{-14}$, $\|\mathfrak{T}_0\| = 1.107$; $k_z = 18$ residual $2.82 \cdot 10^{-14}$, $\|\mathfrak{T}_0\| = 1.250$; $k_z = 42$ residual $3.89 \cdot 10^{-13}$, $\|\mathfrak{T}_0\| = 1.000$; $k_z = 130$ residual $1.03 \cdot 10^{-12}$, $\|\mathfrak{T}_0\| = 1.000$, $\text{ind}_- = 0$.*

The identity $\Pi_K = 0$ (depth $d_0 = |Y|$) reproduces a residual of order the r404 miss: $\|R_{\text{graph}} - R\| = 28.5$, and Euler-on- Y versus A_0 has relative residual 282.

Lemma 4 (Augmented graph factor). *Let $\mathfrak{T}_0$ be CLS at depth $N_w - 1$, vt the image of the genuine border functional λ_B (r362 **bvec_chunked**, not a virtual node), and k the Y -vanishing extra column of CLS with ℓ_B . The min-norm lift $u_{\min} = \mathfrak{T}_0(\mathfrak{T}_0^* \mathfrak{T}_0)^{-1} \text{vt}$ plus the kernel column scaled to the r362 γ produces $\mathfrak{T}^\dagger = [\mathfrak{T}_0 \mid u_{\min} + \alpha k]$ with*

$$(\mathfrak{T}^\dagger)^* \mathfrak{T}^\dagger = (R^\dagger)^{-1} - I$$

to $7.42 \cdot 10^{-14}$ against the r362 Gram, and $R_{\text{graph}}^\dagger = R_{\text{r362}}^\dagger$ to $2.18 \cdot 10^{-14}$. On FRAME-A, $\|\mathfrak{T}^\dagger\|_2 = 0.999917$ ($\lambda_{\min}(R^\dagger) = 0.50004146$). A virtual star-node has spread 56.8 (r405: Sherman–Morrison border $\neq$ aggregated z^{K+1}).

This uses the r362 Gram as the comparison object. It is a dictionary, not a new existence theorem for $R^\dagger \succ \frac{1}{2}I$.

2 What Φ_N is, and what it is not

Proposition 1 ($\Theta_{\text{terminal}} = C^T \Phi_N C$ refuted). *The r369 mixed-Haynsworth matrix $\Phi_N = J^{-1} + U^T A_0^{-1} U$ is the Woodbury precision template of Lemma 2. On FRAME-A its eigenvalues are approximately $(-2.813, -0.0665, 1.804)$. The pair-plus-border Schur complement of the augmented Gram $Z - I$ has eigenvalues approximately $(0.289, 0.294, 0.999)$. Relative mismatch 0.96. The last sequential Uvarov pivot on FRAME-A is $+0.633$, not $-\text{sch}$ and not the middle Haynsworth eigenvalue. Literal $\Theta_{\text{terminal}} = C^T \Phi_N C$ is therefore **refuted**.*

Proposition 2 (Unfitted Krein transport refuted). *Let F be the r404 Euler feature matrix on X and $F^{\text{tr}} = F \cdot \mathfrak{T}_0$. The positive Gram $F^{\text{tr}}(F^{\text{tr}})^T$ has no negative eigenvalue and relative residual 5.62 against $\mathcal{D}_0$. The cos/sin Krein combination $F_+ F_+^T - F_- F_-^T$ after transport has alignment $3 \cdot 10^{-4}$ and residual 3.37. No unfitted scalar recovers $\mathcal{D}_0 = \mathfrak{T}_0$ -transported Euler. A pure positive Gram is excluded.*

3 Census and kills

Sequential Uvarov innovations are the last Schur pivots of the growing- Y family $\mathcal{D}_m$, *not* the LDL of the terminal $\mathcal{D}_0$ (that LDL is Sylvester-tautological with $\text{ind}_- \mathcal{D}_0$). Source order is the grid order of Y ; $|u|$ -descending order kills the FRAME-A count.

Proposition 3 (Innovation census, not a theorem). *On core-42, $n_{\text{neg}}^{\text{inn}} \leq 1$ on 42/42 (0 on 24 windows, 1 on 18). Equality with $\text{ind}_- A_0$ holds on 32/42 only: the ten mismatches $k_z \in \{20, 22, 23, 30, 31, 34, 40, 48, 60, 82\}$ are all of type $0 \neq 1$ (the sequential count undercounts the inertia). Living χ_3 : 37/37 have $\text{ind}_-(I - \mathfrak{T}_0^* \mathfrak{T}_0) \leq 1$; dead χ_3 : 5/5; χ_4 -20: 0. Scramble: algebra residual $1.81 \cdot 10^{-14}$, $\text{ind}_- = 21$, $\|\mathfrak{T}_0\| = 16.66$, sequential count $12 \neq 21$. Weight-permutation: residual $3.34 \cdot 10^{-12}$, $\text{ind}_- = 20$ (kills P1; the graph identity still holds). Gauss weight-jitter: residual $1.64 \cdot 10^{-14}$, ind_- flips $1 \rightarrow 20$. The six terminal-dead χ windows all satisfy $\|\mathfrak{T}^\dagger\| > 1$ (operator flip), including χ_3 -23 and χ_3 -33 whose unaugmented $\mathfrak{T}_0$ stayed contractive; the r405 ones-vector lift stayed contractive on those windows.*

The algebra is world-blind. The *signature* of the innovations carries the arithmetic.

4 What this reduces, and what it does not

P1 on a canonical window is exactly one singular value of $\mathfrak{T}_0$ above 1, equivalently $\text{ind}_-(I - \mathfrak{T}_0^* \mathfrak{T}_0) = 1$, equivalently (r407) $\lambda_2(C) < 1$ fails. That is the same threshold, now with a Jacobi-free source constructor. It does not identify Φ_N with the terminal Uvarov precision, does not transport Euler/Digamma without a fitted Krein form, and does not prove $n_{\text{neg}}^{\text{inn}} = \text{ind}_- A_0$. The augmented identity of Lemma 4 reuses the r362 Gram: it does not close $R^\dagger$ as a new theorem and does not move the mincut.

Machine check

Standalone over $\mathbb{Q}$: three bases, graph = dual CD, kernel dimension, Woodbury $\Theta = 22/29$, and the two named mutants ($u^\vee = 1/u$, $d_0 \pm 1$). Construction: FRAME-A graph versus r407 chain, inertia dictionary, sequential pin, $\Pi_K = 0$ and Euler-direct must-fails, scramble / permute / jitter, Φ_N mismatch, Krein miss, $\mathfrak{T}^\dagger$ versus r362, virtual-node spread, dead- χ flip. Discovery census: `borodin.birkhoff_intertwiner_probe.py` (`--smoke` 31/31 / full 37/37), SPEC.SHA `baee9fc568d8e4ee`, wall 110.4s (builder fallback not taken; bar 120s). Exit line: BORODIN BIRKHOFF VERIFIED.

Claim boundary

Finite identities for the quotient graph, the source CLS formula, the Woodbury precision template, and a named census of sequential Uvarov signs on tested windows. No statement about zeros of $\zeta(s)$, in either direction. No L^* claim. No $R^\dagger \succ \frac{1}{2}I$ claim. No RH claim.

References

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